

20.3.25

Operators in SQL

1. Arithmetic operators

+, -, *, /

2. Concatenation operators

||

3. Comparison operators

=, !=, >, <

4. Relational operators

>, <, >=, <=

5. Logical operators

AND, OR, NOT

6. Special operators

(to simplify the structure of Query)

1. IN

2. NOT IN

3. BETWEEN

4. NOT BETWEEN

5. IS

6. IS NOT

7. LIKE

8. NOT LIKE

7. Subquery operators

1. ALL

2. ANY

3. EXISTS

4. NOT EXISTS

Concatenation operator

- It is used to join the given strings

- The symbol for the concatenation operator is ||

eg1> SELECT 'MR.' || ENAME
FROM EMP;

OUTPUT

'MR.' || ENAME
MR. SMITH
MR. ALLEN

eg2> SELECT 'HII' || ENAME || 'UR SALARY IS' || SAL
FROM EMP;

OUTPUT

'HII' || ENAME || 'UR SALARY IS' || SAL
HII SMITH UR SALARY IS 800
HII ALLEN UR SALARY IS 1600

eg3> SELECT 'HIF' || ENAME || 'UR GF NAME IS MIA'
FROM EMP
WHERE ENAME = 'SMITH';

OUTPUT

HIF SMITH UR GF NAME IS MIA

Logical operator

- It is used to write multiple conditions in a where

Clause

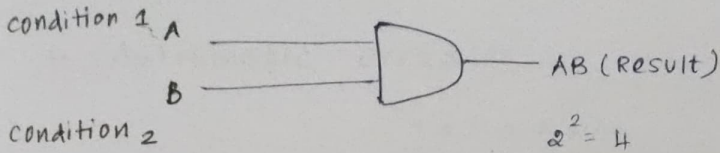
- AND, OR, NOT operator

AND operator

- AND operator is a binary multiplication operator.
- AND operator returns true if all the conditions get

satisfied

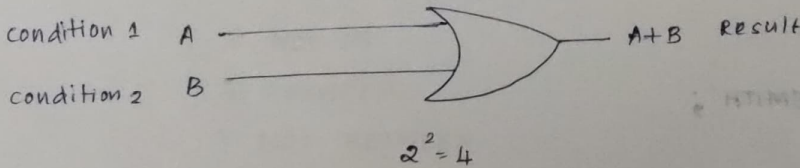
AND (Binary Multiplication)



A	B	AB
T	F	F
F	T	F
F	F	F
T	T	T

OR Operator.

- OR operator is a ~~binary~~ addition operation
- OR operator returns true if any one of the conditions get satisfied.



A	B	A+B
T	F	T
F	T	T
T	T	T
F	F	F

NOT operator

- NOT operator is a inverse operator and it is also known as unary operator (it can accept only one operand) at the right hand side
- NOT operator is used to inverse the results
- We can use NOT operator before the conditions
- NOT operator is similar to not equals operator



A	$\bar{A}$
T	F
F	T

Deptno != 20

Deptno <> 20

NOT Deptno = 20

21-3-25

1. Write a Query to display job and department no. of an employees if the employees are working as a manager in department number 30.

Select job, deptno

from emp

where job = 'MANAGER' AND deptno = 30;

2. write a Query to display details of an employees who are working as a salesman in department no 30 and earning salary greater than 1000.

Select *

from emp

where job = 'SALESMAN' AND deptno = 30 AND sal > 1000;

3. write a Query to display name and sal of an employee who are earning salary greater than 1000 and less than 3000

Select ename, sal

from emp

where sal > 1000 and sal < 3000;

4. WAQTD details of an employee along with annual salary if the employees are working as a analyst and hire before the year 82.

```
select emp.*, sal*12 as "Annual-Sal"
from emp
where job = 'Analyst' And hiredate < '01-JAN-82';
```

5. WAQTD name and job of the employees if the employees are working as a manager or a salesman.

```
select ename, job
from emp
where job = 'MANAGER' or job = 'SALESMAN';
```

6. WAQTD name, salary and department number of the employees if the employees are working in the department of 10, 20, 40

```
select ename, sal, deptno
from emp
where deptno=10 or deptno=20 or deptno=40
```

7. WAQTD details of an employees except who are working in deptno 30.

```
select *
from emp
where not deptno=30;
or
deptno != 30
or
deptno <> 30
```

8. write a query to display name and sal of an employee except who are earning salary < 1000 ;

```
Select ename, sal
from emp
where not sal < 1000 ;
```

sal ≥ 1000

9. write a query to display details of an employees working as a manager in deptno 10 or 20

```
Select *
from emp
where job = 'manager' AND (deptno = 20 or deptno = 10);
```

10. WAQTD name, job, department no. of the employees if the employees are working as a clerk or a manager in dept no

20.

```
Select ename, job, deptno
from emp
where (job = 'CLERK' or job = 'MANAGER') and deptno = 20;
```

NOTE

Whenever using two operation the OR operator should be enclosed in bracket parenthesis.

11. WAQTD details of an employees if the employees are working as clerk or analyst in department number 10 or 20.

```
Select *
from emp
where (job = 'CLERK' or job = 'ANALYST') and (deptno = 10
or deptno = 20);
```


12. WAQTD name and department no of an employees except who are working in deptno 10 or dept no. 20.

```
select ename, deptno
from emp
where not (deptno = 10 or deptno = 20);
```

deptno != 10 and deptno != 20

13. WAQTD details of an employees except who are working as a salesman or analyst.

```
select *
from emp
where job != 'SALESMAN' and job != 'ANALYST';
```

NOT (job = 'SALESMAN' or job = 'ANALYST');

Assignment - 3

1. WAQTD details of the employees working as CLERK and Earning Less than 1500.

```
select *
from emp
where job = 'CLERK' and sal < 1500;
```

2. WAQTD Name and Hiredate of the employees working as manager in dept 30.

```
select ename, hiredate
```

```
from emp
```

```
where job = 'MANAGER' and deptno = 30;
```

3. WAQTD details of the EMP along with annual salary if they are working in dept 30 as salesman and their annual salary has to be greater than 14000.

```
Select EMP.*, sal * 12 as "annual_sal"
```

```
From emp
```

```
Where deptno = 30 and job = 'SALESMAN' and  
sal * 12 > 14000 ;
```

4. WAQTD all the details of the EMP working in dept 30 or as analyst.

```
Select *
```

```
From emp
```

```
Where deptno = 30 or job = 'ANALYST' ;
```

5. WAQTD NAMES of the employees who salary is less than 1100 and their designation is clerk

```
Select ename
```

```
from emp
```

```
Where sal < 1100 and job = 'CLERK' ;
```

6. WAQTD Name and sal, annual sal and deptno if deptno is 20, earning more than 1100 and annual salary exceeds 12000

```
Select ename, sal, sal * 12 as "annual_sal", deptno
```

```
from emp
```

```
Where deptno = 20 and sal > 1100 and sal * 12 > 12000 ;
```

7. WAQTD EMPNO and names of the employees working as manager in dept 20

```
Select empno, ename
```

```
from emp
```

```
Where job = 'MANAGER' and deptno = 20 ;
```


8. WAQTD details of employees working as Analyst in dept 10.

```
select *  
from emp  
where job = 'ANALYST' and deptno = 10 ;
```

9. WAQTD details of employees working in dept 20 or 30.

```
select *  
from emp  
where deptno = 20 or deptno = 30 ;
```

10. WAQTD details of employee working as president with salary of rupees 4000

```
select *  
from emp  
where job = 'PRESIDENT' and sal = 4000 ;
```

11. WAQTD Names and deptno, job of emps working as clerk in dept 10 or 20.

```
select ename, deptno, job
```

```
from emp
```

```
where job = 'CLERK' and (deptno = 10 or deptno = 20)
```

12. WAQTD details of employees working as clerk or manager in dept 10.

```
select *
```

```
from emp
```

```
where (job = 'CLERK' or job = 'MANAGER') and deptno = 10 ;
```

13. WAQTD names of employees working in dept 10, 20, 30, 40

```
select ename
```

```
from emp
```

```
where deptno = 10 or deptno = 20 or deptno = 30 or deptno = 40 ;
```

14. WAQTD details of employees with empno 7902, 7839

select *

from emp

where empno = 7902 or empno = 7839 ;

15. WAQTD details of employees working as manager or salesman or clerk.

select *

from emp

where job = 'MANAGER' or job = 'SALESMAN' or job = 'CLERK' ;

16. WAQTD names of employees hired after 81 and before

87.

select ename

from emp

where hiredate > '31-DEC-81' and hiredate < '01-JAN-87' ;

17. WAQTD details of employees earning more than 1250

but less than 3000.

select *

from emp

where sal > 1250 and sal < 3000 ;

18. WAQTD names of employees hired after 81 into dept 10 or 30.

select ename

from emp

where hiredate > '31-DEC-81' and (deptno = 10 or deptno = 30) ;

19. WAQTD names of employees along with annual salary for the employees working as manager or clerk into dept

10 or 30.

select ename, sal * 12 as "Annual_sal"

from emp

where (job = 'MANAGER' or job = 'CLERK') and (deptno = 10 or

deptno = 30) ;

20) WAQTD all the details along with annual salary if sal is between 1000 and 4000 annual salary more than 15000

```
Select EMP.*, sal*12 as "Annual_sal"
from emp
```

where $sal \geq 1000$ and $sal \leq 4000$ and $sal*12 > 15000$

21) WAQTD details of all the employees except the emps working in dept 10 or 30.

```
Select *
```

```
from emp
```

where $deptno \neq 10$ and $deptno \neq 30$;

22) WAQTD details of all emps along with annual salary except the employees working as ANALYST or MANAGER.

```
Select EMP.*, sal*12 as "Annual_sal"
```

```
from emp
```

where $job \neq 'ANALYST'$ and $job \neq 'MANAGER'$;